Shehjar Sadhu (she/her)

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Research Summary

My research centers on designing Digital health, mHealth platforms that integrate wearable sensors and AI to support remote psycho-physiological health monitoring. My research interests include user-centered design for clinical data analysis platforms for continuous longitudinal monitoring. I have previously worked across diverse health domains, including Attention Deficit Hyperactivity Disorder (ADHD), Parkinson's disease, opioid use disorder, and stress management. Machine Learning Researcher specializing in building scalable pipelines for multimodal clinical data. Expertise in physiological signal processing (ECG, PPG) and deploying mHealth systems on AWS/GCP cloud infrastructure. I have 3+ years of work experience as a Researcher in a Clinical + Technical Lab.

EDUCATION

PhD in Electrical Engineering, *Department of Electrical, Computer, and Biomedical Engineering*

University of Rhode Island (URI), Kingston, RI | Jan 2022 – Present (Tentative Graduation May 2026)
Advisors: Dr. Kunal Mankodiya (Major), Dr. Dhaval Solanki (Co-Major)
Research Areas: Digital Health, Artificial Intelligence (AI), Wearable Sensors

MS in Electrical Engineering, *Department of Electrical, Computer, and Biomedical Engineering*

University of Rhode Island, Kingston, RI | Sep 2020 – Dec 2021
Thesis: CarePortal: A clinician-centered digital health portal for wearable data analytics

BS in Computer Science & Data Science (Dual Degree), *Department of Computer Science and Statistics*

University of Rhode Island, Kingston, RI | Sep 2016 – May 2020

Remote Online Learning - *Coursera Platform | DeepLearning AI | Generative AI with Large Language Models Course | Sep 2025 - Present*

AWARDS AND HONOURS

- **MindGame ADHD Project Featured in Rhody Today**
  News release highlighting project impact | 2025
- **IEEE Body Sensors Network (BSN) Student Travel Award**
  Awarded to attend the international conference twice | 2024, 2023
- **IEEE Connected Health: Applications, Systems, and Engineering Technologies (CHASE) Student Travel Award**
  Awarded to attend the international conference | 2022
- **NSF-NIH Smart Health Workshop Selection**
  Proposal selected for presentation, CareHub project | 2022
- **The University of Rhode Island, Enhancement of Graduate Research Award**
  Smart Diet System for Polycystic Kidney Disorder | 2022
- **IEEE CASS COVID-19 Special Student Design Competition**
  3rd Place – RespDetect: Smart Mask | 2020
- **Undergraduate Research Grant Award, University of Rhode Island**
  Remote Homology Detection [PI: Dr. Noah Daniels] | 2019
- **The University of Rhode Island, Enhancement of Graduate Research Award**
  Samya: A Jewelry-like Wrist Wearable for Managing Stress in Women with Polycystic Ovary Syndrome | 2025

Student Mentoring Achievements

- *(URI)² Undergraduate Grant*, **MindMap** – Aidan Donnellan (CS), 2025
- *(URI)² Undergraduate Grant*, **MindGame** – Elijah Castillo (CS), 2023
- *(URI)² Undergraduate Grant*, **Snapshot Snack** – Savannah Weyhmiller (CS), 2022

PRACTICAL RESEARCH EXPERIENCE, Wearable Biosensing Lab

Ongoing Research

Minder: Cloud System for Wearable Armband to monitor Opioid Use Disorder

Collaboration: UMass Chan Medical School | PhD Work | NIH R01 (Best Demo Award at IEEE BSN 2025)

- **Architected a Serverless Data Pipeline:** Designed and implemented a scalable ETL (Extract, Transform, Load) pipeline on AWS using Lambda functions and S3 buckets to automate the ingestion and synchronization of high-frequency multimodal sensor streams.
- **Multimodal Signal Processing:** Engineered data ingestion workflows for complex physiological time-series data, 6+ hour streaming sessions, specifically Electrocardiogram (arm-ECG), EDA (Electrodermal Activity), and PPG, ensuring data integrity for longitudinal analysis.
- **Full-Stack IoT Integration:** Developed a cross-platform mobile application (Flutter) to interface with custom textile sensors, handling data transmission and cloud synchronization for clinical datasets.
- Currently being used for data collection with N=50 participants by the UMass Chan team and the URI team research assistants to collect, sync, and annotate multimodal sensor data, and is still recruiting.

CareWear: Multimodal Stress Detection Platform for Mental Health

Collaboration: Brown Health | PhD Work | Internal URI funding

- **High-Volume Data Engineering:** Developed a robust Python processing pipeline optimized to handle and clean large-scale physiological datasets (>16GB), integrating streams from commercially available smartwatches and custom chest belts.
- **Automated Feature Extraction:** Implemented advanced signal processing algorithms in Python to denoise and extract clinical biomarkers from ECG (Electrocardiogram), Respiration Rate, and Heart Rate variability (HRV) data. Utilized Pan Tompkins for RR interval detection and Heart Rate extraction.
- **Machine Learning Modeling:** Trained and validated **AI/ML models** for binary and multi-class stress classification, bridging the gap between raw sensor data and clinical insights for physical and mental stress detection.

MindGame: Internet of Medical Things Puzzle Platform for ADHD Behavior Analysis

Collaboration: URI Psychology | PhD Work | Internal URI funding

- **End-to-End System Architecture:** Led the full-stack development of an Internet of Medical Things (IoMT) platform, integrating **backend infrastructure** to securely capture and synchronize interaction metrics (mouse dynamics) with wearable sensor data.
- **Production Deployment & Version Control:** Managed the deployment of the platform for longitudinal data collection, implementing data protocols and backend upgrades to support multi-phase pilot studies (N=4, N=24 participants).
- **Prototyping with EEG:** Currently, engineering a new prototype integration involving the **MUSE EEG headset**, expanding the platform's capacity for neurological signal analysis.
- **R&D work:** Published and submitted research at IEEE BSN and ACM IoT, while securing competitive funding ($30K Faculty Proposal; $1k URI^2), with backend upgrades enabling randomized testing and automated quality assessments.

Past Research

RiseAbove: Epilepsy Self-Management Platform

Collaboration: Brown Health | PhD Work | Epilepsy Foundation, New England

- Collaborated with a neuropsychologist to develop requirements for an online stigma-reduction tool.
- Developed and deployed a containerized Python Flask web application on Google Cloud Platform (GCP), managing server-side logic and database integration for Epilepsy education purposes.

FidgetSense: Hyperactivity Detection in ADHD

Collaboration: Q2BHEAVE LLC | PhD Work |RI State funding & Industry Collaboration

- Collected multimodal smartwatch + video data from N=20 ADHD participants.
- Built ML pipeline for detecting fidgeting/inattention; achieved 90%+ accuracy.
- Deployed real-time behavior detection on Galaxy Watch 4.

Kaya: Smart Glove for Parkinson's Symptom Monitoring

Collaboration: Brown Health | PhD Work | NSF-CAREER grant

- Developed ML models (~90% accuracy) for tremor and rigidity detection using glove sensor data.
- Performed feature engineering and classification using scikit-learn.
- Presented and published at The EEE CHASE conference, MDPI Sensors journal.

CarePortal: Clinician Centered Dashboard

Collaboration: Brown University | Master's Thesis Work | NIH RO1

- Integrated real-time sensor streams; optimized UX based on 21 clinician interviews.
- Sensor data was collected from patients visiting the hospital emergency room who were wearing a commercially available smartwatch.

CareWell: A digital health platform for the education of dementia caregivers

Collaboration: Brown Health | Master's Thesis Work | NIH R21

- Designed and developed a digital health platform containing education content for caregivers of dementia patients.
- Collected data in a feasibility study running for 4 weeks with the app deployed in at-home settings with N=20 caregivers. Presented the platform as a poster in the annual showcase.

Self-Led Research

MultiBallTrack: A Brain training workout game to train attention with RAG implementation for brain workout recommendations

- **Engineered a Retrieval-Augmented Generation (RAG) framework** for a cognitive training application, integrating Google's Gemini LLM with real-time Firestore user telemetry to generate personalized, clinically-grounded neuro-rehabilitation schedules.
- **Developed a cross-platform mobile assessment tool** using Flutter and Firebase, enabling the precise capture of millisecond-level reaction times and attention metrics while implementing scalable cloud architecture for longitudinal behavioral data analysis.
- **Architected robust data fusion pipelines** in Python to process high-frequency biosignal streams, solving critical temporal synchronization challenges between non-aligned sensor arrays and event logs to ensure sub-second ground-truth accuracy for machine learning models

BiosignalViz: Multimodal biosignal processing dashboard with agentic AI recommendations for data processing pipelines for big data.

- Architected BiosignalViz, a high-performance dashboard for visualizing high-volume multimodal biosignals (ECG), utilizing a Firebase backend to log and optimize system performance metrics.
- Engineered an Agentic AI module that autonomously analyzes signal quality and recommends optimal data processing pipelines for cleaning and artifact removal in big data contexts.
- Benchmarked system scalability and rendering latency using the MIT-BIH Arrhythmia Database, ensuring robust performance for clinical-grade datasets.

TEACHING AND OUTREACH ACTIVITIES

Graduate Teaching Assistant, Biomedical Engineering Capstone — Senior Year Undergraduates

University of Rhode Island, Department of Electrical, Computer, and Biomedical Engineering | Sep 2022 – May 2023 | Sep 2023 - May 2024

- Designed student workshops on feature engineering and AI using MATLAB for biomedical signal processing.
- Organized end-of-semester showcase competitions twice annually, with invited external judges.
- Coordinated and hosted graduate school application panels featuring alumni and current graduate students.

Undergraduate Teaching Assistant, *Object-Oriented Programming* — Second Year Undergraduates

University of Rhode Island, Computer Science Department| Sep 2018 – May 2019

- Conducted weekly lab sessions and office hours for C++ programming assignments.
- Assisted students with debugging code and resolving issues across Windows and Mac environments.
- Supported foundational instruction in object-oriented programming concepts and terminal-based development.

Teaching Coach, *Introduction to MicroBit*, K-12 Students

University of Rhode Island Outreach, Computer Science Department | Sep 2018 – May 2019

- Led workshops for K-12 students to learn MicroBit with a step-tracking project using Python.
- Guided students to place sensors on different body parts to collect and interpret biosignal data.
- Emphasized hands-on learning and creativity through end-of-session project demonstrations.

Summer Camp Instructor, *Scratch Programming* — K-12 Students

University of Rhode Island Summer Camp, Computer Science Department | Summer 2018

- Designed and delivered a one-week Scratch programming curriculum for students under 18.
- Provided technical support and real-time debugging for student-created projects.
- Communicated with parents regarding student progress and final project presentations.

INDUSTRIAL WORK EXPERIENCE

Baylor College of Medicine (Voluntary Remote Appointment), Houston, TX

Research Intern | Jun 2023 – Aug 2023

- Deployed a video processing web app on a Linux server with Apache configuration to assess cognitive frailty in older adults.
- Enabled video upload functionality, feature extraction, and CSV export of computed signal metrics.

Embrace Home Loans, Middletown, RI

Capital Markets Intern I | Jul 2020 – Aug 2020

- Built and evaluated regression pipelines (Multiple, Ridge, Lasso, Polynomial) to predict mortgage servicing rights (MSR) valuations.
- Improved model accuracy through hyperparameter tuning and comparative analysis of performance metrics.

FarSounder Inc., Warwick, RI

Software Engineering Intern | Feb 2020 – Mar 2020

- Developed a Python-based analysis tool to compare versions of underwater target detection software.
- Processed and visualized large-scale geospatial datasets using Matplotlib to evaluate sonar accuracy and user impact.

Thermal Solution Resources, Warwick, RI

Software Engineering Intern | Jun 2018 – Aug 2018

- Designed a React-based web interface for configuring Raspberry Pi WiFi credentials via SSID and password input.
- Integrated a MongoDB NoSQL backend to store and manage user configuration data.

TECHNICAL SKILLS

- **Programming & Data Science:**
  Python (NumPy, SciPy, scikit-learn, Matplotlib, Pandas, Seaborn, PyTorch), SQL, R (RStudio), Flask, Hugging Face: LoRA(Low Rank Adaptation), FLAN T5 models, Gemini API usage.
- **Version Control & Collaboration:**
  GitHub, REDCap (survey design), Neptune.ai (version control for AI models)
- **Cloud & Web Infrastructure:**
  Google Cloud Platform (Firebase, Cloud Run), Apache Web Server, Ngrok, AWS (S3, Lambda, SageMaker AI)
- **App Development:**
  Android (Java, Kotlin), Flutter (Dart), Python (Flask).
- **Ethics & Compliance Training:**
  IRB for Full/Expedited protocols (projects: MindGame, CareWear, Kaya, CarePortal, Q2behave)

SELECTED PUBLICATIONS

Journal Publications

[1] **Sadhu** S, Ravichandran V, Bhagat N, Beatty A, Mankodiya K, Weyandt L, Costea G, Solanki D, *FidgetSense: Commodity Smartwatch-Based Monitoring of Fidgeting Behaviors in Children with ADHD.* **ACM Transactions on Computing for Healthcare.** [Impact Factor: 8.0] **[Under Review]**

[2] **Sadhu S,** Solanki D, Constant N, Ravichandran V, Cay G, Saikia MJ, Akbar U, Mankodiya K. *Towards a telehealth infrastructure supported by machine learning on edge/fog for Parkinson's movement screening.* **Smart Health.** 2022 Dec 1;26:100351. [Cite Score: 4.9]

[3] **Sadhu S**, Solanki D, Brick LA, Nugent NR, Mankodiya K. *Designing a Clinician-Centered Wearable Data Dashboard (CarePortal): Participatory Design Study.* **JMIR Formative Research.** 2023 Dec 5;7:e46866. [Impact Factor: 2.2]

[4] Chapman KR, Maynard T, **Sadhu S,** Mankodiya K, Uebelacker L, Davis JD, Ott BR, Tremont G. *Beta Test of a Multicomponent Mobile Health Application for Dementia Caregivers.* **Journal of Technology in Behavioral Science.** 2023 Jul 7:1-0. [Impact Factor: 2.3]

[5] Ravichandran V, **Sadhu S**, Convey D, Guerrier S, Chomal S, Dupre AM, Akbar U, Solanki D, Mankodiya K. iTex Gloves: *Design and In-Home Evaluation of an E-Textile Glove System for Tele-Assessment of Parkinson's Disease.* **Sensors.** 2023 Mar 7;23(6):2877. [Impact Factor: 3.576]

[6] Cay G, Ravichandran V, **Sadhu S**, Zisk AH, Salisbury AL, Solanki D, Mankodiya K. *Recent advancement in sleep technologies: A literature review on clinical standards, sensors, apps, and AI methods.* **IEEE Access.** 2022 Sep 28;10:104737-56. [Impact Factor: 3.9]

[7] Prieto S, Kiriakopoulos ET, Goldstein A, Kaden S, Tremont G, Mankodiya K, Castillo E, **Sadhu S,** Solanki D, Davis JD, Margolis SA. Toward a multimodal model of internalized epilepsy stigma. **Epilepsy & Behavior**. 2026 Jan 1;174:110812. [Impact Factor: 2.3]

[8] Prieto S, Kiriakopoulos ET, Goldstein A, Kaden S, Tremont G, Mankodiya K, Castillo E, **Sadhu S,** Solanki D, Davis JD, Margolis SA. Stigma intersectionality and its impact on an epilepsy stigma self-management program. **Epilepsy & Behavior**. 2025 Nov 1;172:110682. [Impact Factor: 2.3]

[9] Margolis SA, Prieto S, Goldstein A, Kaden S, Castillo E, **Sadhu S**, Solanki D, Larracey ET, Tremont G, Mankodiya K, Kiriakopoulos ET. Feasibility and acceptability of an online epilepsy stigma self-management program. **Epilepsy & Behavior**. 2025 Apr 1;165:110331. [Impact Factor: 2.3]

[10] **Sadhu S**, Solanki D, Mankodiya K, MA al Rumon. *CareWear: A multimodal physiological dataset collected via a consumer-based wearable device for stress monitoring.* (Target: Nature Scientific Data, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies) **[Under Preparation]**

Conference Proceedings

[1] **Sadhu S,** Bhagat N, Castillo E, Weyandt L, Mankodiya K, and Solanki S. 2025. *Is wearable data reliable for monitoring behavior? Design of a wearable-based IoMT puzzle game for remote behavior monitoring.* In **The 15th International Conference on the Internet of Things (IOT 2025)**, November 18–21, 2025, Vienna, Austria. ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3770501.3770523

[2] **Sadhu S**, Castillo E, Weyandt L, Solanki D, Mankodiya K. *Feasibility of a Digital Health Puzzle Game for Detecting Computer Mouse Behavioral Patterns in ADHD.* In 2024, **IEEE 20th International Conference on Body Sensor Networks (BSN)** 2024 Oct 15 (pp. 1-4). IEEE.

[3] Hicking F, **Sadhu S**, Ravichandran V, Weyandt L, Costea GO, Mankodiya K, Solanki D. *Comparative Investigation of Smartwatch Data in Children with ADHD and Non-ADHD.* **In 2024 IEEE 20th International Conference on Body Sensor Networks (BSN),** 2024 Oct 15 (pp. 1-4). IEEE.

[4] **S. Sadhu**, V. Ravichandran, N. Constant, U. Akbar, K. Mankodiya, D. Solanki. *Exploring the Impact of Parkinson's Medication Intake on Motor Exams Performed in-home Using Smart Gloves*. 2023 **IEEE 19th International Conference on Body Sensor Networks (BSN)**.

[5] **S. Sadhu,** A. Yerule, N. Constant, U. Akbar, and K. Mankodiya, "*Motor exercise classification using machine learning.*" **2019 IEEE MIT Undergraduate Research Technology Conference (URTC)**, 2019, pp. 1-4.

[6] Seckin M, **Sadhu S**, Al Rumon MA, Gravel M, DiFazio H, Johnson N, Perry K, Solanki D, Mankodiya K. MedDock: A 3D-Printed Smart Pill Dispenser with Sensitive Textile Sensor for Adherence Monitoring. **In 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), 2024** Oct 15 (pp. 1-4). IEEE.

JOURNAL REVIEW SERVICES

- PLOS ONE journal
- IEEE Journal of Biomedical Health Informatics.
- Springer Nature: Biomedical Materials & Devices.
- JMIR Human Factors.

STUDENT MENTORING AND COLABORATION EXPERIENCE

Biomedical Engineering Capstone Mentoring (Undergraduate)

- Mentored two teams of 4 students to synthesize a research problem into defined project scopes along with timelines, with 1 year to finish the project related to diet monitoring.
- Advised on experimental design, data analysis, or technical implementation. Making sure the project was delivered on time.

- Providing regular feedback on troubleshooting, project reports, posters, demos, and showcase preparation.
- Projects included electrical circuit design, 3D printing, and human study (IRB) design to test the system.

WBL Research Assistants mentoring

- Mentored undergraduate students supported by NSF and NIH projects, and created teams to work together on project requirements.
- Provided mentorship on statistical analysis of human subject data using R and Python app development.
- Provided support and mentorship for applying to undergraduate grants. Received three grants for undergraduate support.

Undergraduate Student Mentees'	Graduation Date
1. Daniel Sanguino, BS Computer Science, URI	May 2026
2. Elijah Castillo, BS Computer Science, URI	May 2026
3. Aidan Donnellan, Computer Engineering, URI	May 2026
4. Sarah Cilento, BS Molecular Neuroscience , URI	May 2024
5. Savannah Weyhmiller, BS Computer Science, URI	May 2023
6. Daniela Acarapi, BS Data Science,	May 2023
7. Tyler Hanlon, BS Computer Science, URI	May 2023
8. Denno Louis, BS in Computer Science	May 2023
9. Danica Kate Ramos, BS Biomedical and Mechanical Engineering, URI	May 2022
10. Jason Stoltz, BS in Biomedical Engineering and BA in Biology, URI	May 2022
11. Lauren Wells, BS in Biomedical Engineering, URI	May 2022

Masters Student Mentee

1. Friederike Hicking, MS in Electrical Engineering, URI	July 2023